Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Moments

MARKSCHEME ($g = 9.8\,\text{m}\,\text{s}^{-2}$)

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1. $M = Fd = 5$ **M1A1** [2]
2. $M = 8 \times 2 \times \sin 30° = 8$ **M1A1** [4]
3. $M = 5g \times 2 = 98$ **M1A1** [3]
4. Each reaction $= 5g = 49$ **M1A1** [3]
5. Moments: $F \times 2 = 20 \times 1 \Rightarrow F = 10$ **M1A1** [3]
6. Balance: $2x = 4(3 - x) \Rightarrow x = 2$ from 2 kg **M1A1** [3]
7. Moments about one end; solve two reactions **M1A1** [5]
8. $M = 12 \times 0.8 = 9.6$ **M1A1** [2]
9. Moments about A: $T \times 2\sin 30° = 6g \times 2 \Rightarrow T = 117.6$ **M1A1** [5]
10. Resolve + moments or Lami/3-4-5; $T_1 = 8$, $T_2 = 6$ **M1A1** [5]
11. Moments about ends; T_1, T_2 **M1A1** [5]
12. $M = 15 \times 2\sin 60° = 15\sqrt{3}$ **M1A1** [3]
13. Moments: $F \times 3\cos 40° = 4g \times 1.5\cos 40° \Rightarrow F = 19.6$ **M1A1** [5]
14. Each $= 2.5g = 24.5$ **M1A1** [3]
15. $\sum \mathbf{F} = \mathbf{0}$ and $\sum M = 0$ **B1** [2]
16. Resolve horiz/vert; moments about base; $R_w = W\cot 60°$, $R_g = W$ **M1A1** [6]
17. Moments about A; geometry of triangle **M1A1** [6]
18. Moments: reactions from $R_1 \times 4 = 10g \times 2.5$ etc. **M1A1** [5]
19. Moments about base including person; $F = \mu R$ check **M1A1** [7]
20. Moments about hinge; $T \times 2a\cos 30° = mg \times a\cos 30°$ **M1A1** [6]
21. $\bar{x} = \frac{0\cdot3+2\cdot2+5\cdot5}{10} = 2.9$ **M1A1** [4]
22. Moments about hinge side **M1A1** [6]
23. $30 \times 1.5 = 25x \Rightarrow x = 1.8$ **M1A1** [4]
24. Moments + resolve; hinge reaction components **M1A1** [5]
25. F required $= \frac{1}{2}W\cot 60°$; compare μR **M1A1** [6]